

EPM213s - Electrical Measurements

Report 4: Range Extension of Moving Iron Voltmeter

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1. Objective

The aim of this experiment is to compute the value of a series multiplier resistance (RsmultiplierR_s^{\text{multiplier}}Rsmultiplier) that must be added to a moving iron instrument in order to expand its voltage measurement range by a specific factor MMM.

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2. **Theoretical Background**:

To extend the voltage range of a voltmeter, a non-inductive high resistance, known as a multiplier, is connected in series with the instrument. Let: * Vm be the original full-scale deflection voltage of the instrument (this corresponds to the voltage across the meter impedance Zm at full scale current). * V_extended be the new desired full-scale deflection voltage. * M be the voltage multiplication factor, such that V_extended = M * Vm. * |Zm| be the internal impedance of the moving iron instrument (as determined in Report 3). * Rs_multiplier be the resistance of the series multiplier. When the multiplier is connected, the same full-scale deflection current (Ifsd) flows through both the meter and the multiplier. For the original range: Vm = Ifsd * |Zm| For the extended range: V_extended = Ifsd * (|Zm| + Rs_multiplier) Substituting V_extended = M * Vm: M * Vm = Ifsd * (|Zm| + Rs_multiplier) M * (Ifsd * |Zm|) = Ifsd * (|Zm| + Rs_multiplier) Dividing by Ifsd (assuming Ifsd ≠ 0): M * |Zm| = |Zm| + Rs_multiplier Therefore, the series multiplier resistance can be calculated as: Rs_multiplier = M * |Zm| |Zm| Rs_multiplier = |Zm| * (M - 1)

3. **Given Data and Previously Determined Values • •**

Voltage Range Multiplication Factor (M): 4 Internal AC Impedance of the Moving Iron Instrument (|Zm|): From Report 3, |Zm| = 161.29 Ohms. ◦ Note: The calculation of inductance (Lm) in Report 3 was indeterminate due to |Zm| < Rm. However, the value of |Zm| itself was determined from the AC experimental data and will be used here as requested for the range extension calculation

4. Calculation of Series Multiplier Resistance:

(Rs_multiplier) Using the formula Rs_multiplier = |Zm| * (M - 1): Rs_multiplier = 161.29 Ohms * (4 - 1) Rs_multiplier = 161.29 Ohms * 3 Rs_multiplier = 483.87 Ohms (Note: The Python script output was 483.86 Ohms, the slight difference is due to rounding in this manual calculation step. The script value is more precise.) Calculated value from script: Rs_multiplier = 483.86 Ohms

5. Conclusion

To increase the voltage measurement range of the given moving iron voltmeter by a factor of **4**, a **non-inductive series resistance of approximately 483.86 Ohms** is required.

This resistor must be non-inductive to maintain measurement accuracy, especially when measuring AC voltages across varying frequencies, and to prevent introducing phase errors into the system.

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6. Discussion

The accuracy of the voltage range extension depends significantly on two factors:

- The precision of the instrument's internal impedance (|Zm||Z_m||Zm|)
- The accuracy of the multiplier resistance (RsmultiplierR_s^{\text{multiplier}}Rsmultiplier)

A concern was identified in Report 3, where the AC impedance |Zm||Z_m||Zm| was found to be less than the DC resistance RmR_mRm. This discrepancy could affect the accuracy of the multiplier calculation, especially if |Zm||Z_m||Zm| does not reliably represent the true AC behavior of the instrument under standard operating conditions.

For more reliable instrument adaptation, a deeper investigation into this mismatch between |Zm||Z_m||Zm| and RmR_mRm is recommended. This would help ensure that the multiplier resistor provides the exact voltage range extension intended.